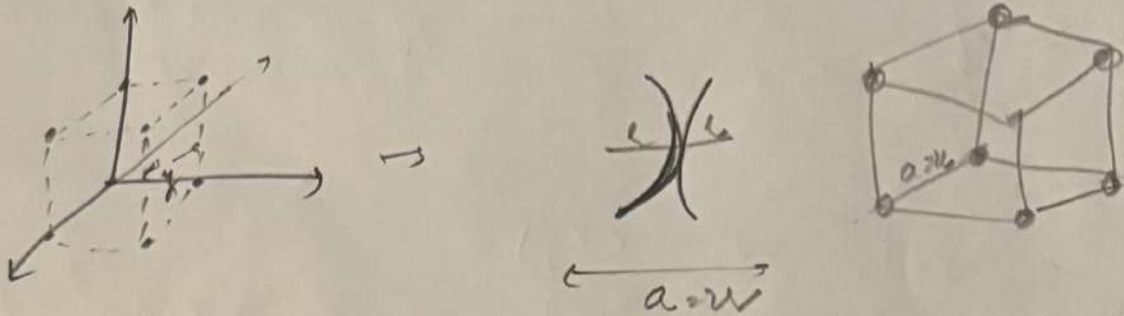


1. Simple Cubic (SC) - touching direction: $\langle 100 \rangle$ (cube edge)

Atoms sit at the 8 corners of the cube. Along a cube edge, one corner atom touches the adjacent corner atom directly.

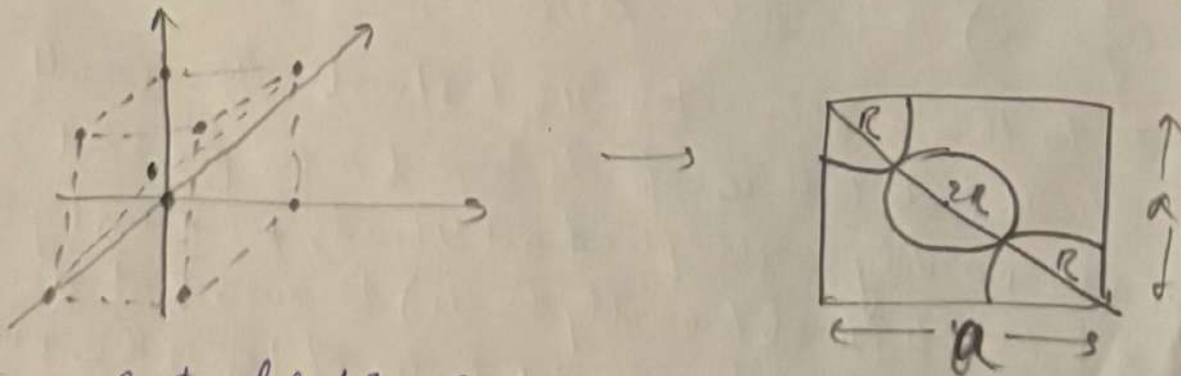
The edge length equals the sum of the two atomic radii:

$$a = r + r = 2r$$

Body Centred Cubic (BCC) - touching direction: $\langle 111 \rangle$ (body diagonal)

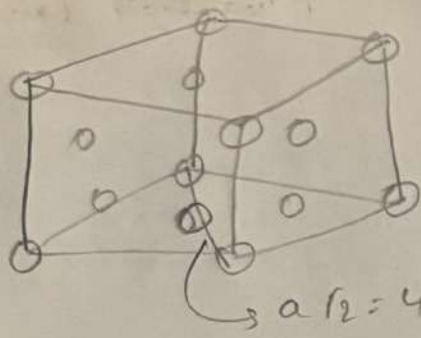
Atoms sit at 8 corners + 1 body centre. Travelling along the body diagonal from one corner to the opposite corner, you pass through: the corner atom (radius r), then the full diameter of the body centred atom ($2r$), then the opposite corner atom (radius r). The body diagonal has geometric length $a\sqrt{3}$, so:

$$a\sqrt{3} = r + 2r + r = 4r \Rightarrow a = 4r/\sqrt{3}$$

Face Centred Cubic (FCC): touching direction: $\langle 110 \rangle$ (face diagonal)

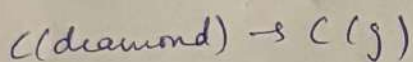
Atoms sit at 8 corners + 6 face centres. On any cube face, travelling along the face diagonal, you pass through: corner atom (r), the full diameter of the face centred atom ($2r$) and the opposite corner atom (r). The face diagonal has geometric length $a/\sqrt{2}$, so:

$$a/\sqrt{2} = r + 2r + r = 4r \Rightarrow a = 2\sqrt{2}r$$



2. Enthalpy of atomisation is the energy required to convert 1 mol of substance into 1 mol of isolated gaseous atoms.

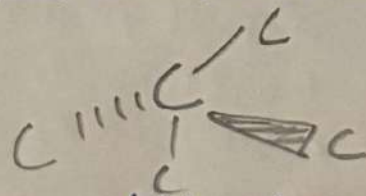
for diamond:



here for 1 mol of diamond Δ atomisation C(diamond) is 713 kJ/mol

Each carbon atom is tetrahedrally bonded to 4 neighbouring carbons. However every bond is shared between the two atoms it connects. so bond belongs half to each atom

$$\text{bonds per atom} = 4 \times \frac{1}{2} = 2$$



The enthalpy needed to atomise one mol of diamond (break every C-atom fully here) is therefore: $\Delta H(\text{atomization}) = 2 \times 347 = 694 \text{ kJ/mol}$

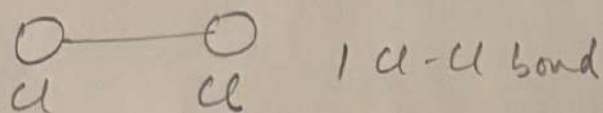
This is close to the experimental 713 kJ/mol $\rightarrow$ the small deviation reflects real lattice effects beyond a simple bond-counting model. If we naively multiplied all 4 bonds per atom by 347 kJ/mol we would get 1388 kJ/mol (exactly double) because that double counts each bond. The bond sharing factor of $1/2$ is the origin of the 'factor of two'.

Cross check using Cl_2

Breaking one Cl-Cl bond in a mole of Cl_2 requires the full bond energy, 242.4 kJ defined per mole of bonds broken. But breaking that one bond produces 2 mole of Cl atoms. So the energy expressed per mol of Cl atom produced:

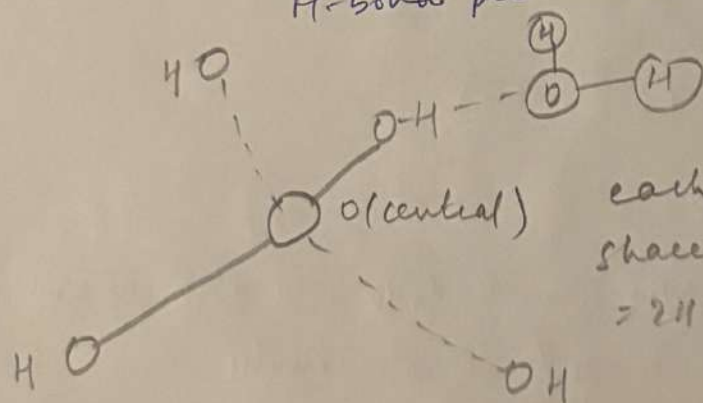
$$242.4 / 2 = 121.2 \approx 121.3 \text{ kJ/mol}^{-1}$$

exactly the given heat of dissociation value. This confirms the general principle: whether we count energy per mole of bonds or per mole of atoms changes the number by factor of 2, because one bond always connects (or when broken releases) two atoms.



3. In ice each H_2O molecule is tetrahedrally hydrogenbonded to 4 neighbouring water molecules $\rightarrow$ 2 bonds where it donates an H, 2 where it accepts. As with diamond, each H-bond is shared between 2 molecules

$$\text{H-bond per molecule} = 4/2 = 2$$



each O has 4 H-bonds (2 donor, 2 acceptor) shared with neighbours $\rightarrow 4/2 = 2$ H-bonds per molecule

Using a typical H-bond energy from textbook, $E(\text{H-bond}) \approx 20 \text{ kJ/mol}$

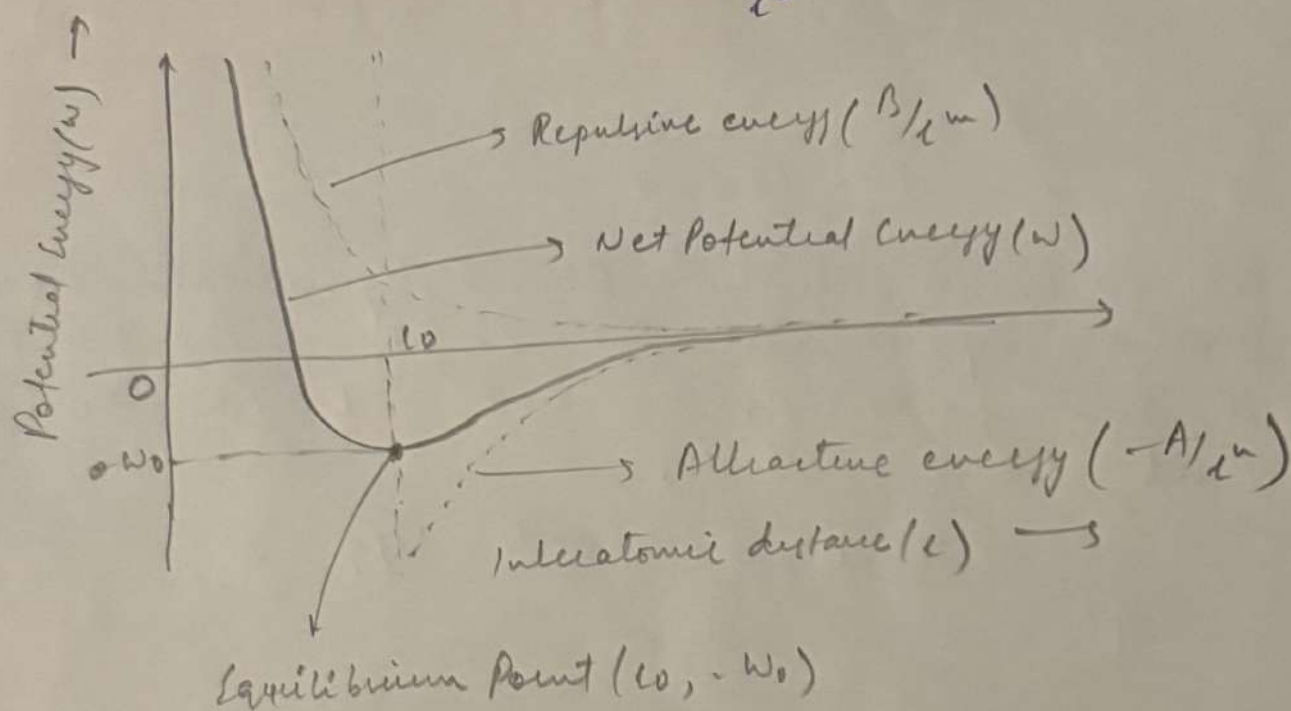
$$\text{Total H-bond energy per mole of water} = 2 \times 20 = 40 \text{ kJ/mol}$$

$$\text{Enthalpy of fusion of ice: } \Delta H(\text{fus}) = 6.01 \text{ kJ/mol}$$

$$\text{fraction disrupted} = \Delta H(\text{fus}) / (2 \times E(\text{H-bond})) = 6.01 / 40 \approx 0.15 \quad (15\%)$$

Melting only disrupts roughly 10-15% of hydrogen bonds. The H-bond network is not destroyed on melting, most H-bonds survive, and liquid water retains substantial short range tetrahedral H-bonded structure (this is why water keeps many ice like properties - high boiling point, high heat capacity, and density maximum near 4°C just above melting point). What's lost on melting is mainly the long range crystalline order/rigidity, not the bonding network itself.

4. $w = -\frac{A}{r^n} + \frac{B}{r^m}$ here w : bond potential energy
 $-\frac{A}{r^n}$ = attractive PE
 $\frac{B}{r^m}$ = repulsive PE



Interatomic Potential energy Well

as $r \rightarrow \infty$ the interaction b/w them tends to zero also $m > n$ and
 when $r \rightarrow 0$ the repulsive force dominates
 And at $r = r_0$ we have ^{minimum} net force b/w them so

$$\left. \frac{dw}{dr} \right|_{r_0} = 0$$

i) At eq^m, net force is zero, i.e. $dw/dr = 0$

$$\frac{dw}{dr} = \frac{nA}{r^{n+1}} - \frac{mB}{r^{m+1}} = 0$$

$$\Rightarrow \frac{nA}{r^{n+1}} = \frac{mB}{r^{m+1}} \Rightarrow n \cdot A \cdot r^{(m-n)} = mB$$

$$\Rightarrow r^{m-n} = mB/nA \Rightarrow \boxed{r_0 = \left(\frac{mB}{nA} \right)^{\frac{1}{m-n}}}$$

ii) from eq^m condition above, we have

$$\frac{m B}{l_0^{m+1}} = \frac{n A}{l_0^{n+1}}$$

$$\frac{A}{l_0^n} = \frac{m}{n} \frac{B}{l_0^m}$$

$$E_{att} = \frac{m}{n} E_{rep}$$

iii) to prove $W_0 = \frac{A}{l_0^n} \left(\frac{m-n}{m} \right)$

$$W = -\frac{A}{l_0^n} + \frac{B}{l_0^m}$$

from above $\frac{B}{l_0^m} = \frac{n}{m} \left(\frac{A}{l_0^n} \right) \Rightarrow W_0 = -\frac{A}{l_0^n} + \frac{n}{m} \left(\frac{A}{l_0^n} \right)$

$$W_0 = -\frac{A}{l_0^n} \left(1 - \frac{n}{m} \right) = -\frac{A}{l_0^n} \left(\frac{m-n}{m} \right)$$

Taking the magnitude of this well depth (the -ve sign just indicates a bound / attractive state):

$$\boxed{W_0 = \frac{A}{l_0^n} \left(\frac{m-n}{m} \right)} \rightarrow \text{proved}$$

5. $W = \underbrace{-\frac{A' z_1 z_2 e^2}{l}}_{\text{attraction}} + \underbrace{\frac{B}{l^m}}_{\text{repulsion}} + D E$ (energy released in bond formation)

Given $l_0 = 2.50 \text{ \AA}$, $m = 9$

$$W = -\frac{C}{l} + \frac{B}{l^m} + D E$$

here $C = \underline{A' z_1 z_2 e^2}$

$$\text{At } e^m \frac{dw}{dr} = 0$$

$$\frac{dw}{dr} = \frac{C}{r^2} - \frac{mB}{r^{m+1}} = 0$$

$$\frac{C}{r^2} = \frac{mB}{r^{m+1}} \Rightarrow B = \frac{r^{m-1} A' z_1 z_2 e^2}{m}$$

$$\Rightarrow B = \frac{e^8 A' (1)(1) e^2}{9} \quad (z_1 = z_2 = 1 \Rightarrow \text{univalent ionic pair})$$

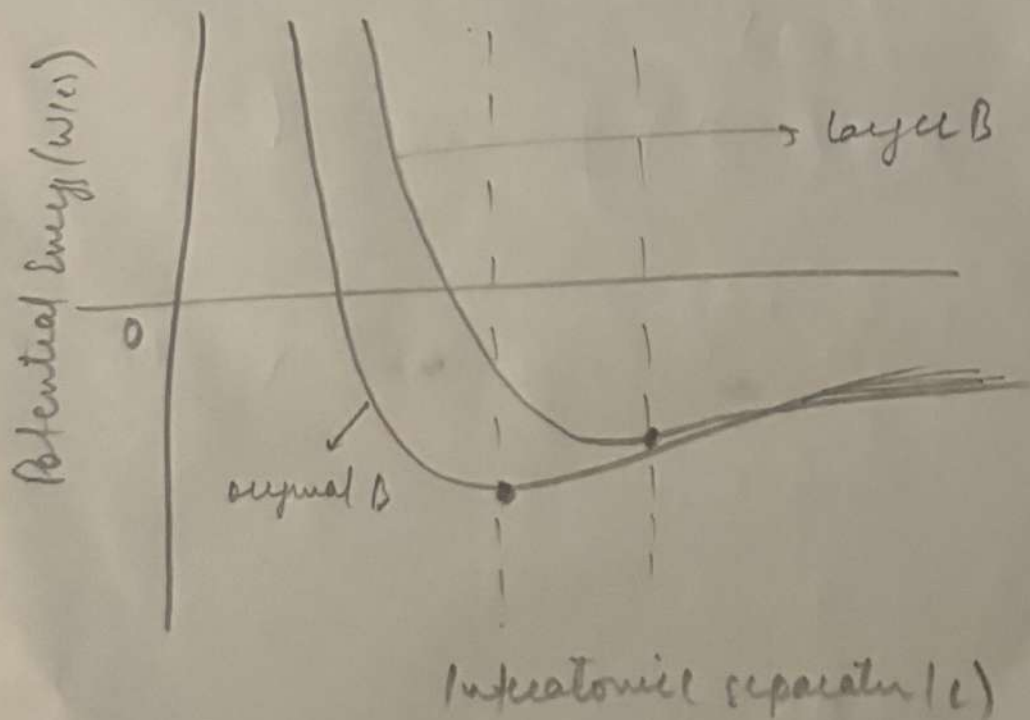
$$\boxed{B = \frac{A' e^2 (2.5)^8}{9}}$$

11) what if B increases

$$r_0 = \left(\frac{mB}{A} \right)^{\frac{1}{m-1}} \Rightarrow r_0 \propto B^{\frac{1}{m-1}}$$

so $r_0 \uparrow$ when $B \uparrow$

$\Rightarrow$ as B increases, r_0 increases, repulsive (force) of potential becomes stronger & pushes back harder at short range so ions settle into e^m at larger separation



Increasing B pushes r_0 to a larger separation

6. Enthalpy of atomisation = No. of bonds per atom $\times$ bond energy.

for Cu, it has FCC crystal structure which has 12 atoms surrounding each atom. Bond per atom = $\frac{12}{2} = 6$ bond/atom

for W, it has BCC crystal structure which has coordination no. of 8 $\rightarrow$ Bond per atom = $\frac{8}{2} = 4$ bond/atom

from table: $\Delta H_{\text{atomisation}}$ for Cu crystal = $6 \times 56.4 = 338.4 \text{ kJ/mol}$

$$\Delta H(W) = 4 \times 212.3 = 849.2 \text{ kJ/mol}$$

Even though Cu has high coordination no. which result in high no. of bonds, the individual bond energy of Cu-Cu is very less than W-W bond which makes the $\Delta H_{\text{atom}}(W) > \Delta H_{\text{atom}}(Cu)$

Tungsten individual bond is nearly 4x stronger than copper's (212.3 vs $56.4 \text{ kJ/mol per bond}$) - this is because W has many more unpaired d-electrons available for strong, covalent like metallic bonding (it sits in middle of the transition series where d-orbital bonding is maximised) given very strong interatomic attraction. This large per-bond strength advantage completely overwhelms the disadvantage of BCC's lower coordination no. (4 vs Cu's 6 bonds/atom). This is consistent with tungsten's extremely highly melting point ($\sim 3400^\circ\text{C}$) compared to copper's (1085°C).
atomisation enthalpy correlates strongly with melting point

7. We need to read the coordinates directly from the figure and then convert them into Miller indices

a) Direction: The direction e starts at the origin and points to the points whose coordinates are $(a, b, c) = (1, 1, 0)$

So the direction components are $\Delta a : \Delta b : \Delta c = 1 : 1 : 0$

therefore, $e = [110]$

$$l : [1, 1, 0]$$

2. Miller indices of the plane indicated by S

The solid plane passes through origin, so we cannot directly take its intercepts. Instead draw a parallel plane that does not pass through the origin. This is the dashed plane shown on right from figure its intercepts are

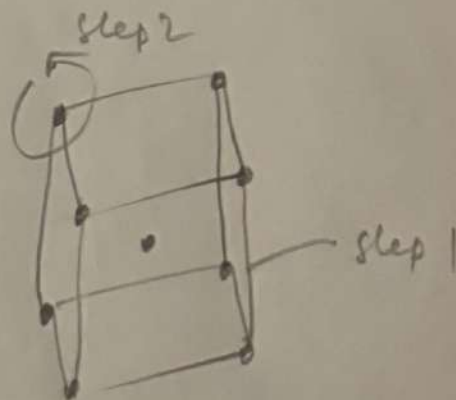
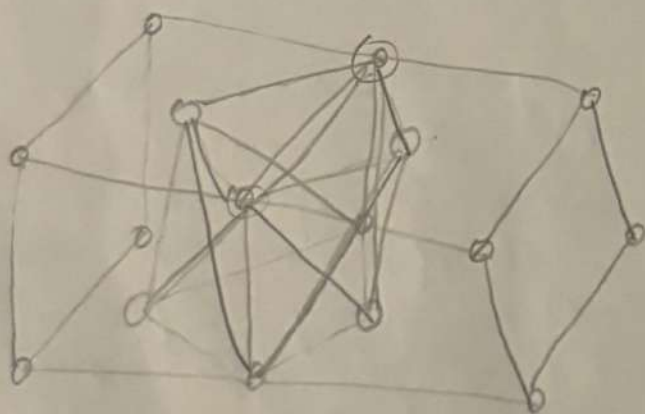
$a=3$, $b=2$, $c=\infty$ (because plane parallel to c-axis & never intersects the c-axis)

Take reciprocal

$\frac{1}{3}$, $\frac{1}{2}$, 0

Plane indicated by S: $(2, 3, 0)$

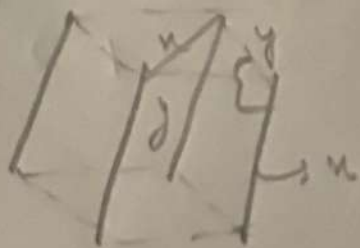
8. i)



It is a tetragonal crystal type as

$$a = b \neq c \quad \& \quad \alpha = \beta = \gamma = 90^\circ$$

The cell outlined is a body-centered tetragonal (BCT) cell. There is central atom on the unit cell and ~~and~~ also there is 8 atoms at the 8 corners.



$$x = 2y \Rightarrow y = \frac{x}{2}$$

So we have 2 sided equal but one side not equal

11) Yes - body centered tetragonal is one of the 14 Bravais lattices

- iii) General BCT lattice has independent axes $a = b \neq c$ (tetragonal, symmetry, any c/a ratio). When you construct the BCT cell from two adjacent FCC cells as described, algebra shows the resulting cell has a specific fixed axial ratio.

$$\frac{c}{a} = \sqrt{2} \quad \left(\frac{x}{y} = \sqrt{2} \right)$$

At exactly this ratio, lower symmetry tetragonal cell accidentally gains extra symmetry elements and becomes indistinguishable in overall lattice symmetry from a cubic cell i.e. it becomes equivalent to FCC.

- Every FCC lattice can be re-described as a BCT cell with $\frac{x}{y} = \sqrt{2}$, but a general BCT lattice (arbitrary x/y) is not FCC. This is why FCC with its higher symmetry is treated as the special simpler case, while BCT with $x/y = \sqrt{2}$ is the more general description of the same underlying lattice.

10. These are van der Waals-bonded (London dispersion forces). Two difficult physical properties are responsible for the two trends, so they aren't contradictory:

- Bond length is governed mainly by atomic size. Going down the group (He $\rightarrow$ Xe) each element has more electron shells, so the atom's electron cloud (and its van der Waals radius) is physically larger. Bigger atoms can't approach each other as closely so the equilibrium separation increases.
- Bond energy is governed mainly by polarizability. Larger atoms have more electrons that are more loosely held, making it easier to distort the electron cloud. This gives stronger instantaneous-dipole/induced-dipole (London dispersion) attractions between neighbouring atoms, hence higher bond energy also why melting/boiling points rise steadily down the group.

So both trends increase with atomic no., for the same underlying reason (more trends electrons $\rightarrow$ bigger, more polarizable electron cloud) but they govern different aspects of the bond, its geometric length versus its strength - so there is no real contradiction b/w them.

11. Crystal follows SC and the weight of 1 mol atom is 74.5 g
Also atomic size is 0.145 nm

By SC $\rightarrow$ we know that edge length of cell is equal to $2R$

so $a = 2R$ $a = 2(0.145) = 0.290 \text{ nm} = 0.290 \times 10^{-9} \text{ m}$

Atoms per unit cell (SC: 8 corner $\times 1/8$ each)

$N = 1 \text{ atom/cell}$

Volume of cell $= a^3 = (0.29 \times 10^{-9})^3 \text{ m}^3$

$V_c = 2.439 \times 10^{-23} \text{ cm}^3$

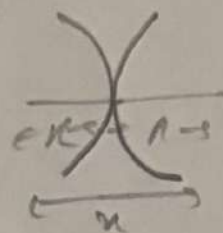
Mass of one cell = 74.5 g $\rightarrow$ 1 g $\rightarrow$ N_A . no. of atoms
 $n \rightarrow$ 1 atom

$n = \frac{74.5}{N_A} = \frac{74.5 \text{ g} \times \text{atom} \times 1 \text{ mol}}{6.022 \times 10^{23} \text{ atom/mol}}$

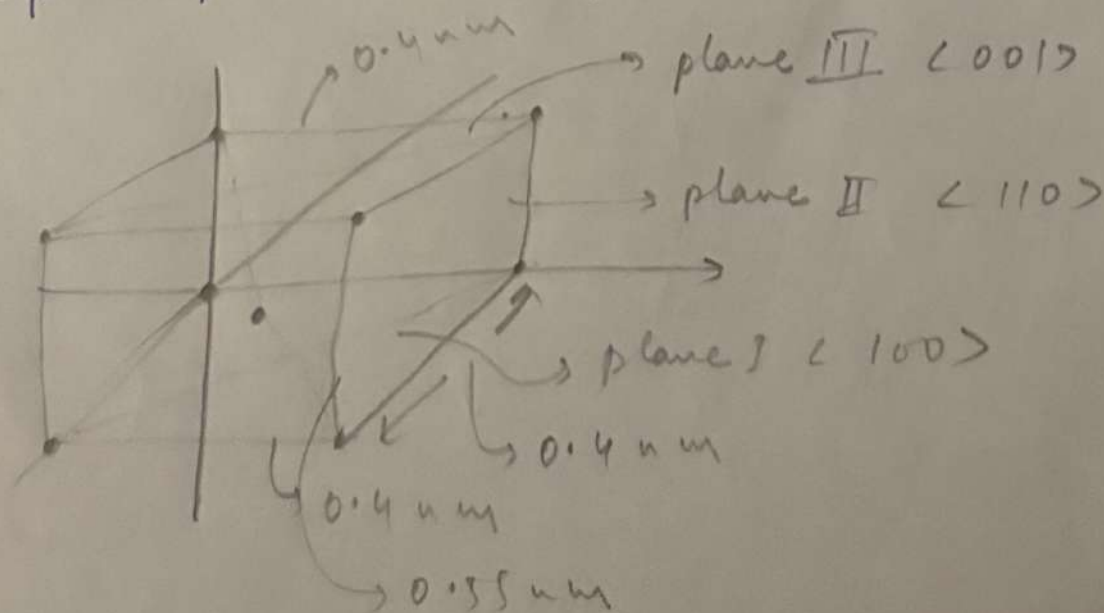
$m = 1.237 \times 10^{-22} \text{ g}$

density $\rho = \text{mass of unit cell} / \text{Vol. of unit cell}$

$= \frac{1.237 \times 10^{-22}}{2.4389 \times 10^{-23}} = 5.07 \text{ g/cm}^3$



12. Given, 3 different views of the unit cell now let's reconstitute this separate plane to one single unit cell



a) from (001) $a=b=0.4\text{nm}$ from (100): the other edge $c=0.55\text{nm}$
 All angles shown are 90° since $a=b \neq c$ and all angles are $=90^\circ$, this is a tetragonal system.

b) The (110) view shows an extra atom sitting in centre of the rectangle (not just at corners) this is body centred atom viewed edge-on. Combined with tetragonal cell shape, this identifies the structure as body-centred tetragonal (BCT)

c) Given density $\rho = 18.9\text{g/cm}^3$

Atoms per cell cent (BCT) $\Rightarrow n = 8 \times \frac{1}{8} (\text{corners}) + 1 (\text{body centre})$
 $= 2 \text{ atoms/cell}$

Cell Volume ($a=0.4\text{nm} = 4 \times 10^{-8}\text{cm}$, $c=0.55\text{nm} = 5.5 \times 10^{-8}\text{cm}$)

$$V_c = a^2 c = (4 \times 10^{-8})^2 (5.5 \times 10^{-8}) = 8.8 \times 10^{-23}$$

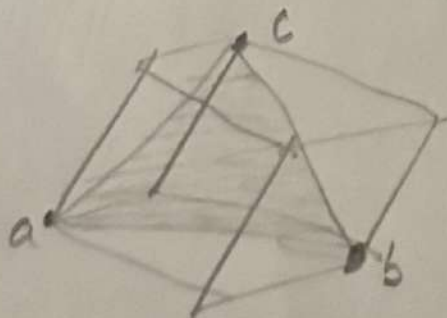
Atomic Weight: $A_w = \frac{\rho \cdot V_c \cdot N_A}{n} = \frac{18.91 \times 8.8 \times 10^{-23} \times 6.022 \times 10^{23}}{2}$

$$\underline{A_w = 801 \text{ g/mol}}$$

9. Triclinic lattice - (111) plane

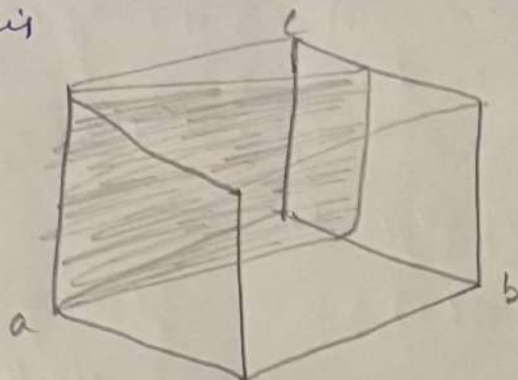
Triclinic $\rightarrow a \neq b \neq c$, $\angle \neq 90^\circ$ (a general skewed parallelepiped)

The 111 plane intercepts each axis at 1 unit from the origin
 mark points at $1/a, 1/b, 1/c$ along 3 non diagonal orthogonal, unequal length-axes, then connect the three intercept points to form triangular plane



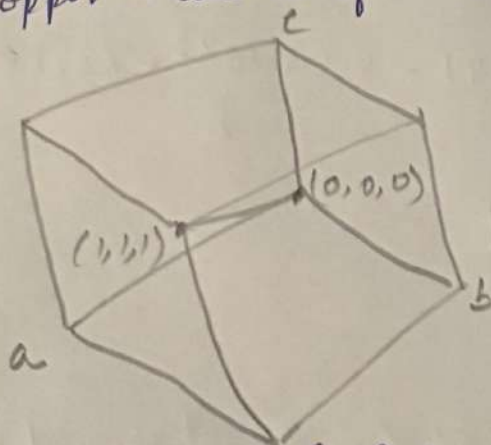
11) Tetragonal lattice - (120) plane

$a = b \neq c$ all angles are 90° , intercepts from (hkl) = $1/20 : 1/1 : 1$
 along a, $1/2 = 0.5$ along b, $1/0 = \infty$ along c (i.e. parallel to c-axis)
 This plane is a vertical sheet thru $1a$ and $0.5b$ running parallel to c-axis



111) Orthorhombic lattice - [111] direction

$a \neq b \neq c$ all angles 90° , single arrow from origin $(0,0,0)$ directly to opposite corner of the box, at $(1a, 1b, 1c)$



111) Cubic lattice - (211) direction

$a = b = c$ indices (211) means vector components are in ratio $2:1:1$
 Since a component (2) is larger than one cell edge, vector is drawn extending across 2 cubes stacked along a, ending at units along a, 1 along b, 1 unit along c

